

I tools di Active Directory

Ci sono diversi strumenti installati sul server per la gestione di AD

- Active Directory Domains and Trusts
- Active Directory Users and Computers
- Active Directory Sites and Services
- Group Policy Management
- DNS
- Active Directory Schema

Questi strumenti sono raggiungibili dal menu «Administrative Tools»

Gli stessi strumenti, per la gestione remota dei Servizi AD, possono essere installati su un normale PC Windows joinato al dominio utilizzando i RSAT (Remote Server Administration Tools)

I tools di Active Directory

The screenshot displays the Windows Server Manager interface. On the left, the 'Dashboard' sidebar shows 'Local Server', 'All Servers', 'AD DS', 'DNS', and 'File and Storage Services'. The main area is titled 'WELCOME TO SERVER M...' and contains a 'QUICK START' section, a 'WHAT'S NEW' section, and a 'ROLES AND SERVER GRO...' section. The 'ROLES AND SERVER GRO...' section shows 'AD DS' as the selected role, with sub-items: 'Manageability', 'Events', 'Services', 'Performance', and 'BPA results'. The 'Administrative Tools' window is open, showing a list of tools. The list includes 'Terminal Services', 'Active Directory Administrative Center', 'Active Directory Domains and Trusts', 'Active Directory Module for Windows PowerShell', 'Active Directory Sites and Services', 'Active Directory Users and Computers', 'ADSI Edit', 'Component Services', 'Computer Management', 'Defragment and Optimize Drives', 'Disk Cleanup', 'DNS', 'Event Viewer', 'Group Policy Management', 'iSCSI Initiator', 'Local Security Policy', 'Microsoft Azure Services', 'ODBC Data Sources (32-bit)', 'ODBC Data Sources (64-bit)', 'Performance Monitor', 'Print Management', 'Recovery Drive', 'Registry Editor', 'Resource Monitor', 'Server Manager', 'Services', 'System Configuration', 'System Information', 'Task Scheduler', 'Windows Defender Firewall with Advanced Security', 'Windows Memory Diagnostic', and 'Windows Server Backup'. The status bar at the bottom indicates '32 items' and '6 items selected 6,95 KB'.

Server Manager Dashboard

WELCOME TO SERVER M...

QUICK START

WHAT'S NEW

LEARN MORE

ROLES AND SERVER GRO...

Roles: 3 | Server groups: 1

AD DS

Manageability

Events

Services

Performance

BPA results

Administrative Tools

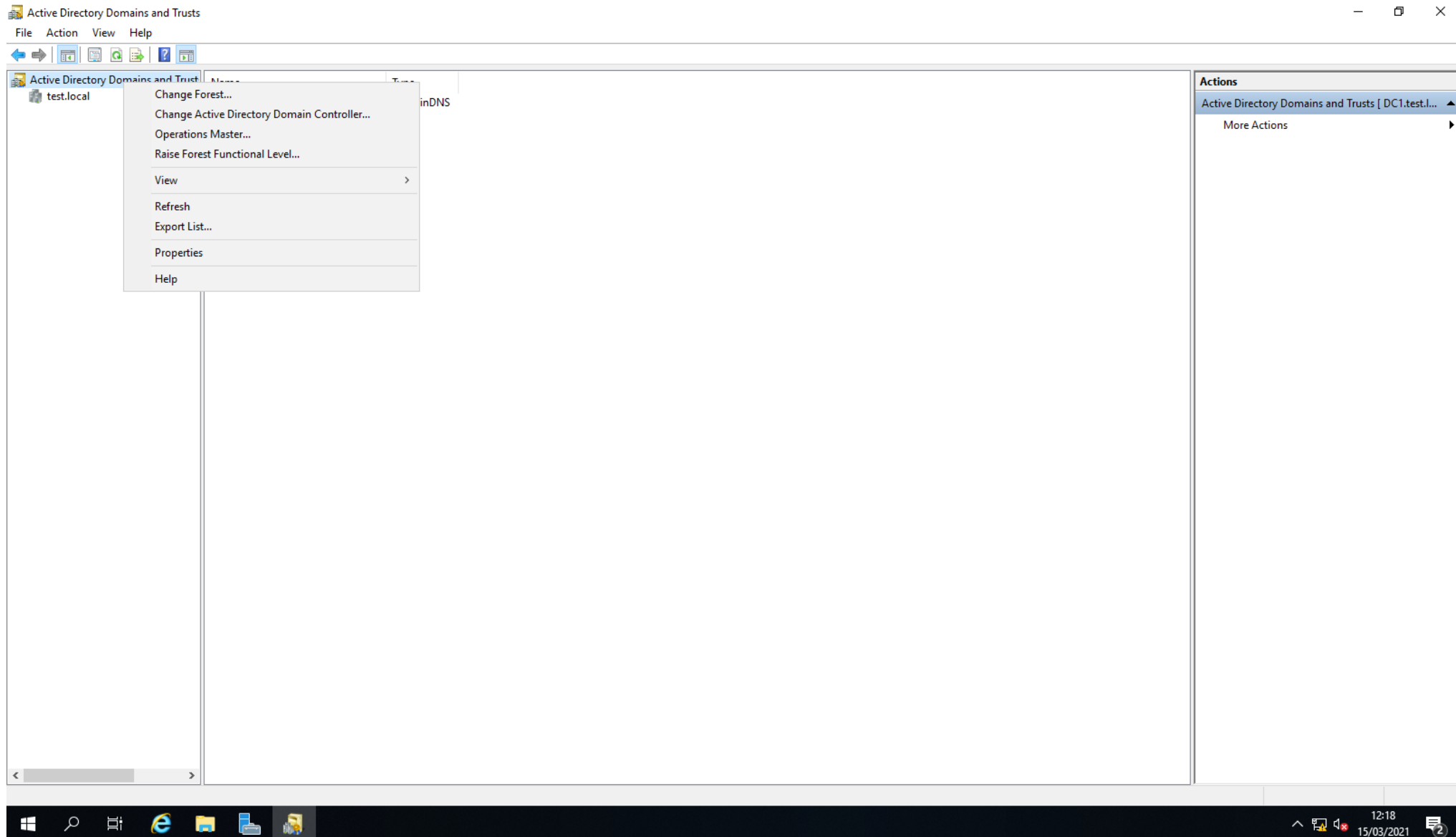
Name	Date modified	Type	Size
Terminal Services	15/09/2018 09:19	File folder	
Active Directory Administrative Center	15/09/2018 09:13	Shortcut	2 KB
Active Directory Domains and Trusts	15/09/2018 09:14	Shortcut	2 KB
Active Directory Module for Windows PowerShell	15/09/2018 09:13	Shortcut	2 KB
Active Directory Sites and Services	15/09/2018 09:13	Shortcut	2 KB
Active Directory Users and Computers	15/09/2018 09:14	Shortcut	2 KB
ADSI Edit	15/09/2018 09:13	Shortcut	2 KB
Component Services	15/09/2018 09:12	Shortcut	2 KB
Computer Management	15/09/2018 09:12	Shortcut	2 KB
Defragment and Optimize Drives	15/09/2018 09:12	Shortcut	2 KB
Disk Cleanup	15/09/2018 09:12	Shortcut	2 KB
DNS	15/09/2018 09:13	Shortcut	2 KB
Event Viewer	15/09/2018 09:12	Shortcut	2 KB
Group Policy Management	15/09/2018 09:13	Shortcut	2 KB
iSCSI Initiator	15/09/2018 09:12	Shortcut	2 KB
Local Security Policy	15/09/2018 09:13	Shortcut	2 KB
Microsoft Azure Services	15/09/2018 09:13	Shortcut	2 KB
ODBC Data Sources (32-bit)	15/09/2018 09:12	Shortcut	2 KB
ODBC Data Sources (64-bit)	15/09/2018 09:12	Shortcut	2 KB
Performance Monitor	15/09/2018 09:12	Shortcut	2 KB
Print Management	15/09/2018 09:13	Shortcut	2 KB
Recovery Drive	15/09/2018 09:12	Shortcut	2 KB
Registry Editor	15/09/2018 09:12	Shortcut	2 KB
Resource Monitor	15/09/2018 09:12	Shortcut	2 KB
Server Manager	15/09/2018 09:13	Shortcut	2 KB
Services	15/09/2018 09:12	Shortcut	2 KB
System Configuration	15/09/2018 09:12	Shortcut	2 KB
System Information	15/09/2018 09:13	Shortcut	2 KB
Task Scheduler	15/09/2018 09:12	Shortcut	2 KB
Windows Defender Firewall with Advanced Security	15/09/2018 09:12	Shortcut	2 KB
Windows Memory Diagnostic	15/09/2018 09:13	Shortcut	2 KB
Windows Server Backup	15/09/2018 09:14	Shortcut	2 KB

32 items 6 items selected 6,95 KB

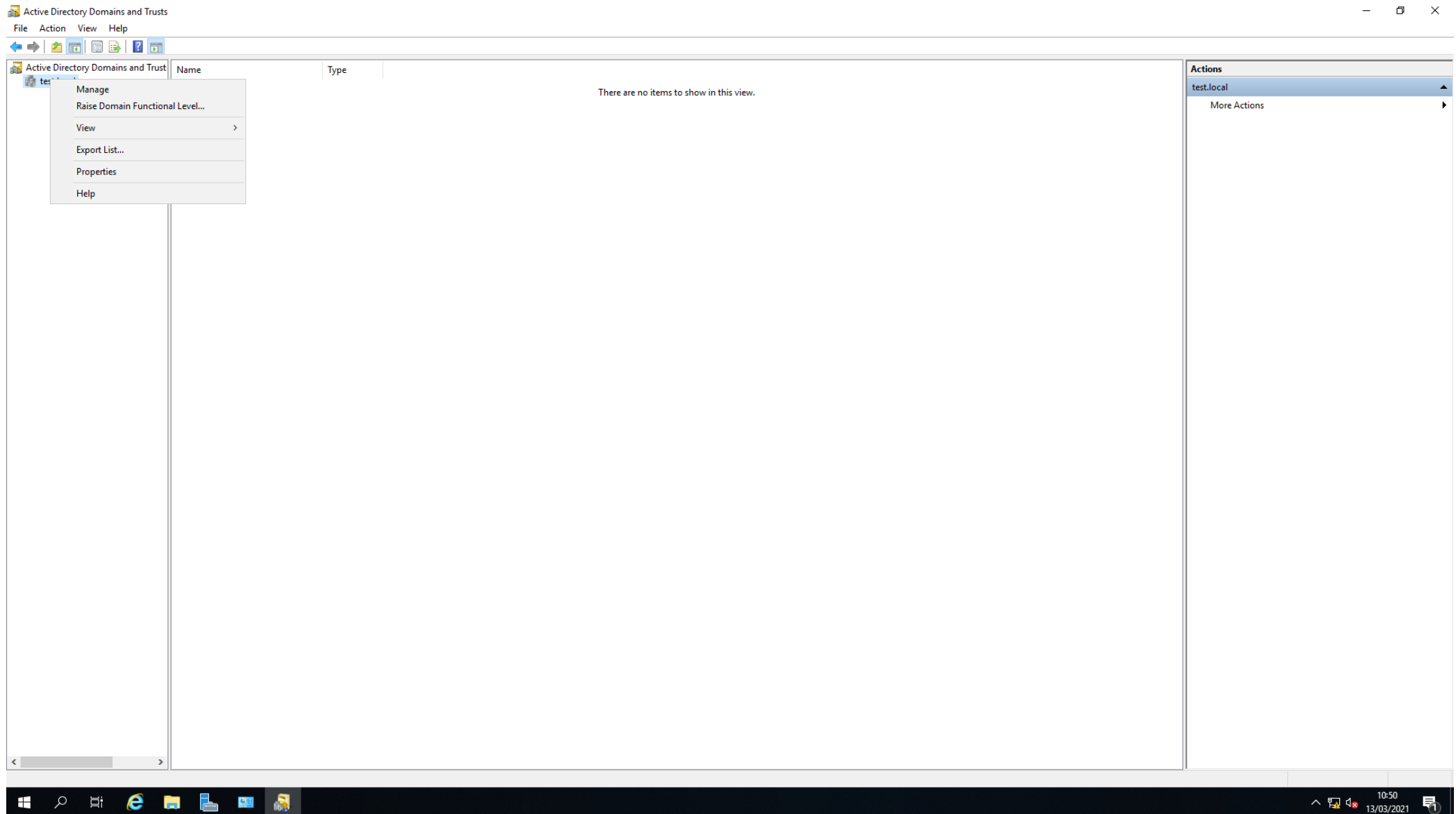
Active Directory Domains and Trusts

- Vediamo i domini che fanno parte della foresta
- Gestione livello funzionalità della foresta
- Gestione livello funzionalità del dominio
- Gestiamo i trust fra domini e foreste
- Cambiamo il ruolo Domain Name Master

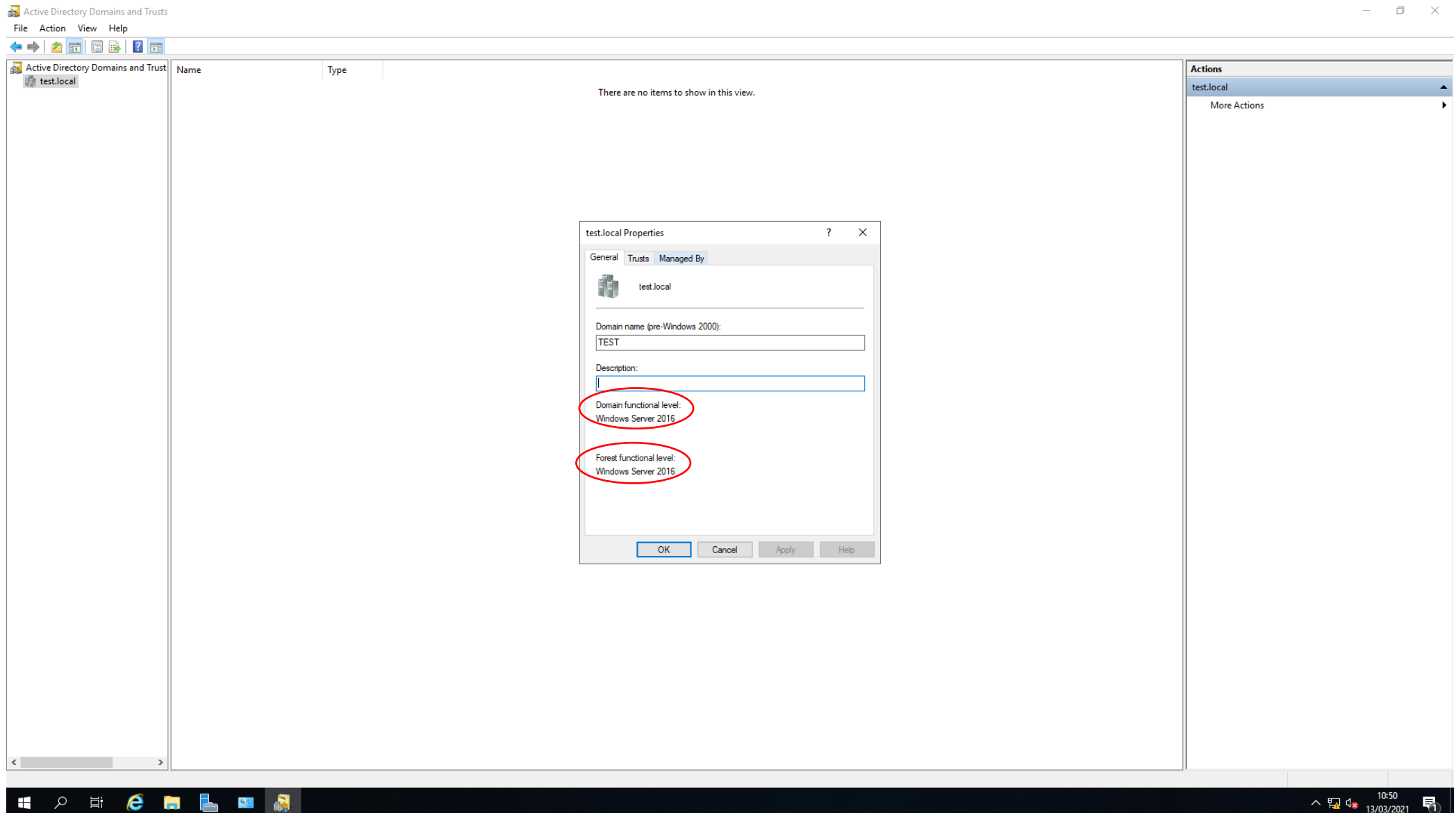
Active Directory Domains and Trusts



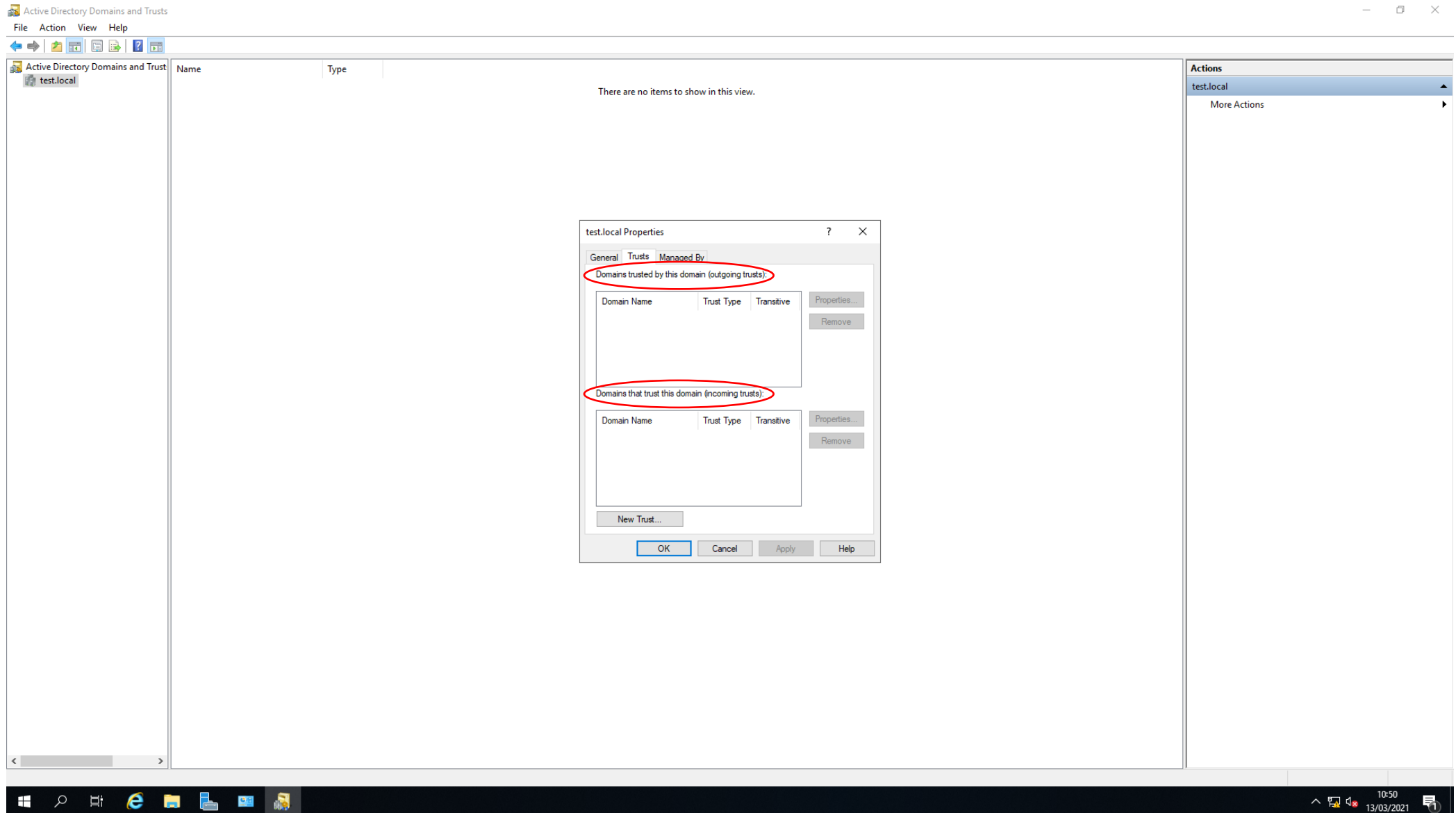
Active Directory Domains and Trusts



Active Directory Domains and Trusts



Active Directory Domains and Trusts



Active Directory Domains and Trusts

The screenshot displays the 'Active Directory Domains and Trusts' console window. The left pane shows a tree view with 'dir.unibo.it' expanded, listing its subdomains: campusfc.dir.unibo.it, hq.dir.unibo.it, personale.dir.unibo.it, poliorimini.dir.unibo.it, and studenti.dir.unibo.it. The main pane shows a table of these domains, all of type 'domainDNS'. A 'dir.unibo.it Properties' dialog box is open, showing the 'Trusts' tab. It lists outgoing trusts (Child, Transitive) and incoming trusts (Child, Forest). The 'hospital.aosp.bo.it' domain is highlighted in the incoming trusts list.

Active Directory Domains and Trusts

File Action View Help

Active Directory Domains and Trusts

dir.unibo.it

- campusfc.dir.unibo.it
- hq.dir.unibo.it
- personale.dir.unibo.it
- poliorimini.dir.unibo.it
- studenti.dir.unibo.it

Name	Type
campusfc.dir.unibo.it	domainDNS
hq.dir.unibo.it	domainDNS
personale.dir.unibo.it	domainDNS
poliorimini.dir.unibo.it	domainDNS
studenti.dir.unibo.it	domainDNS

dir.unibo.it Properties

General Trusts Managed By

Domains trusted by this domain (outgoing trusts):

Domain Name	Trust Type	Transitive
campusfc.dir.unibo.it	Child	Yes
hq.dir.unibo.it	Child	Yes
personale.dir.unibo.it	Child	Yes
poliorimini.dir.unibo.it	Child	Yes
studenti.dir.unibo.it	Child	Yes

Domains that trust this domain (incoming trusts):

Domain Name	Trust Type	Transitive
campusfc.dir.unibo.it	Child	Yes
hospital.aosp.bo.it	Forest	Yes
hq.dir.unibo.it	Child	Yes
personale.dir.unibo.it	Child	Yes
poliorimini.dir.unibo.it	Child	Yes

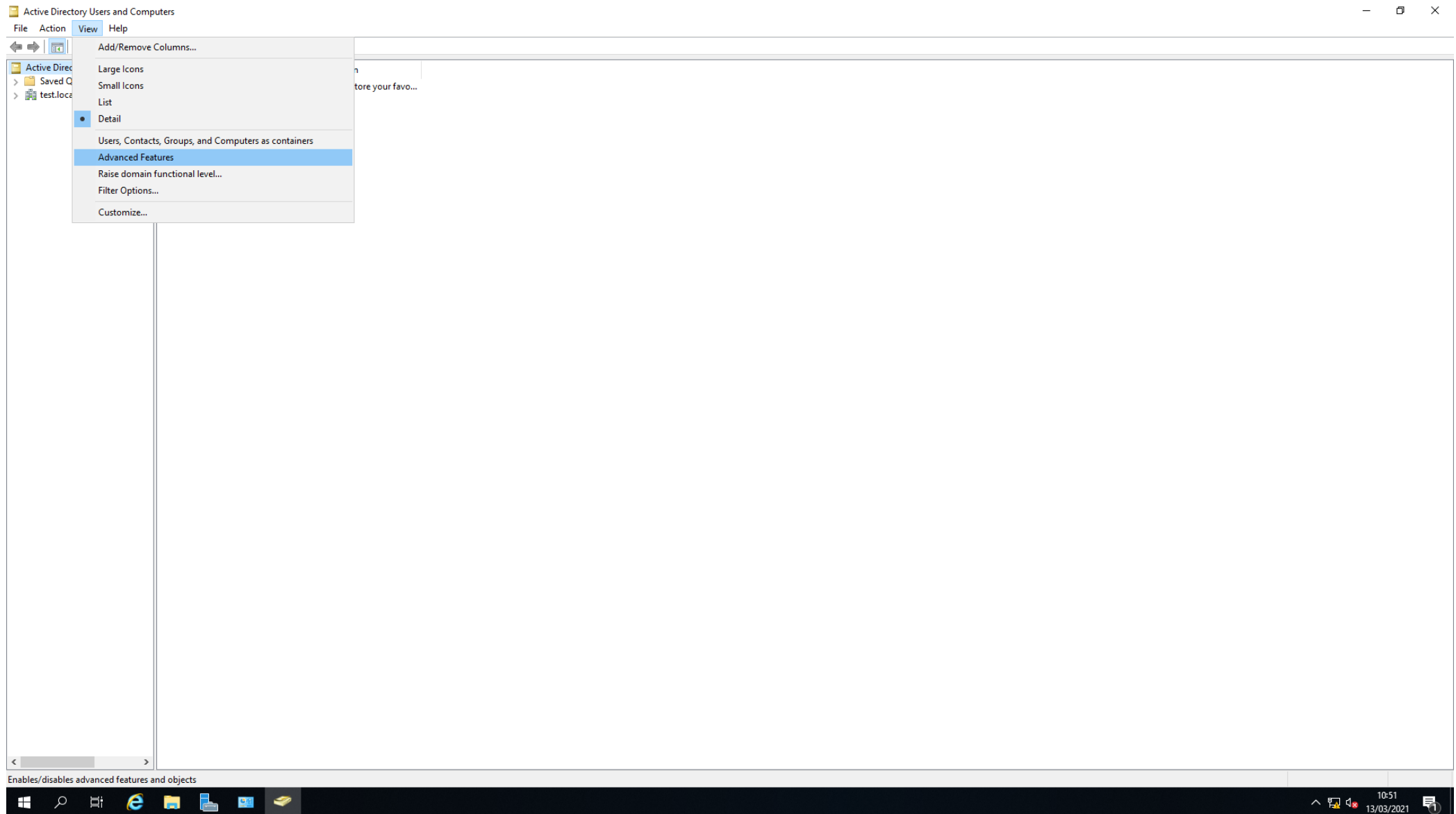
OK Cancel Apply Help

Windows taskbar: 12:26 PM 3/15/2021

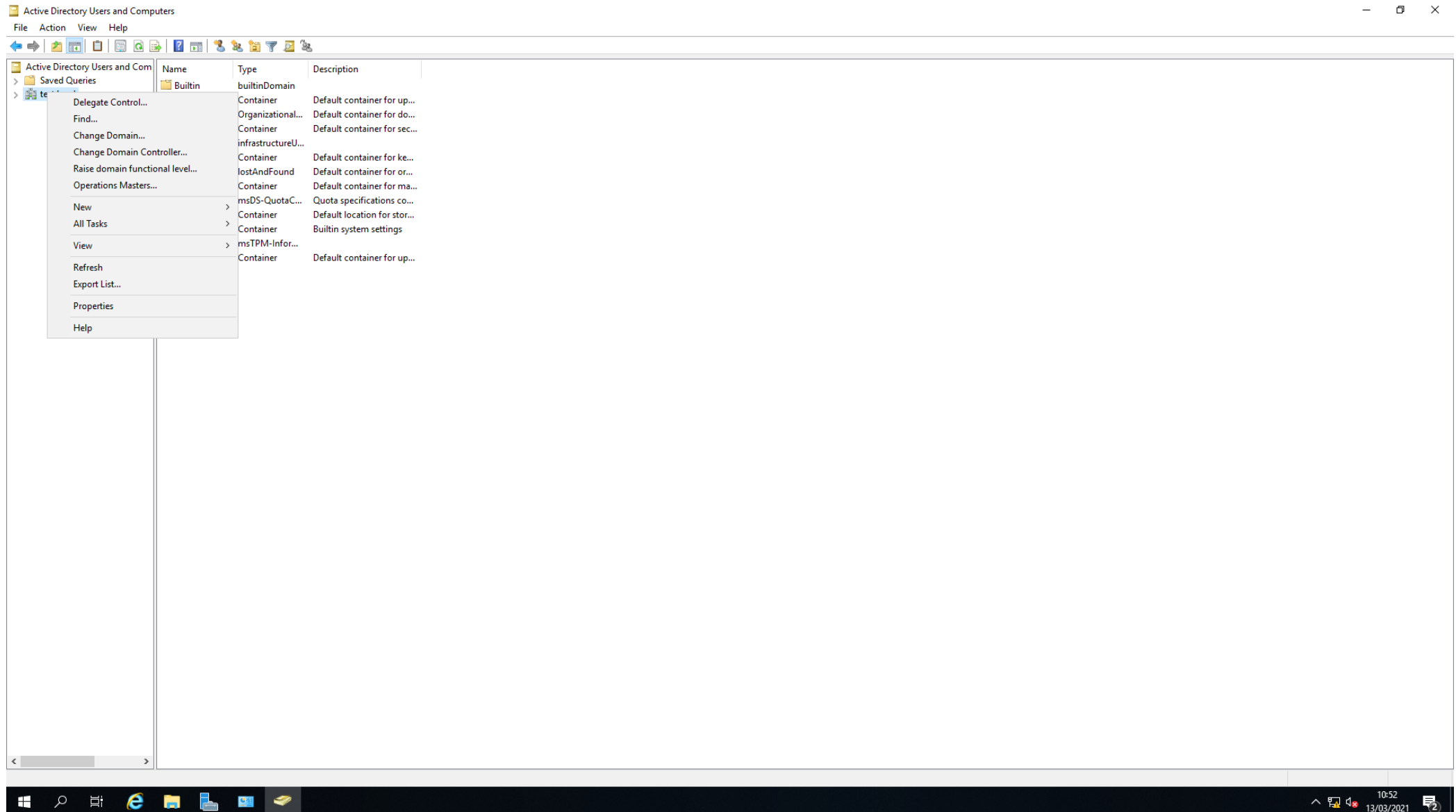
Active Directory Users and Computers

- Gestiamo utenti, gruppi e computer
- Creiamo le OU
- Cambiamo il livello di funzionalità del dominio
- Assegniamo le deleghe
- Cambiamo i ruoli di PDC, RID e infrastructure master

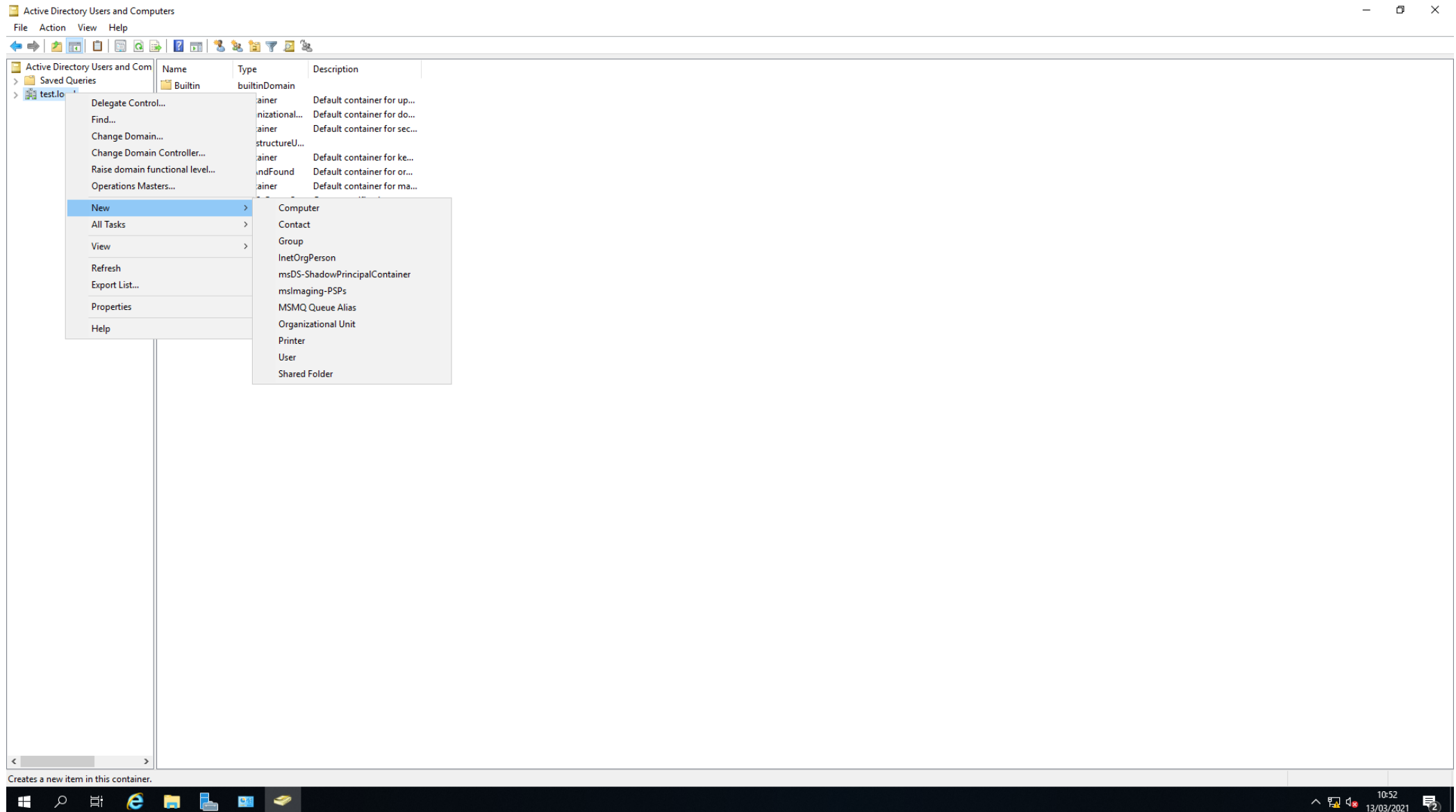
Active Directory Users and Computers



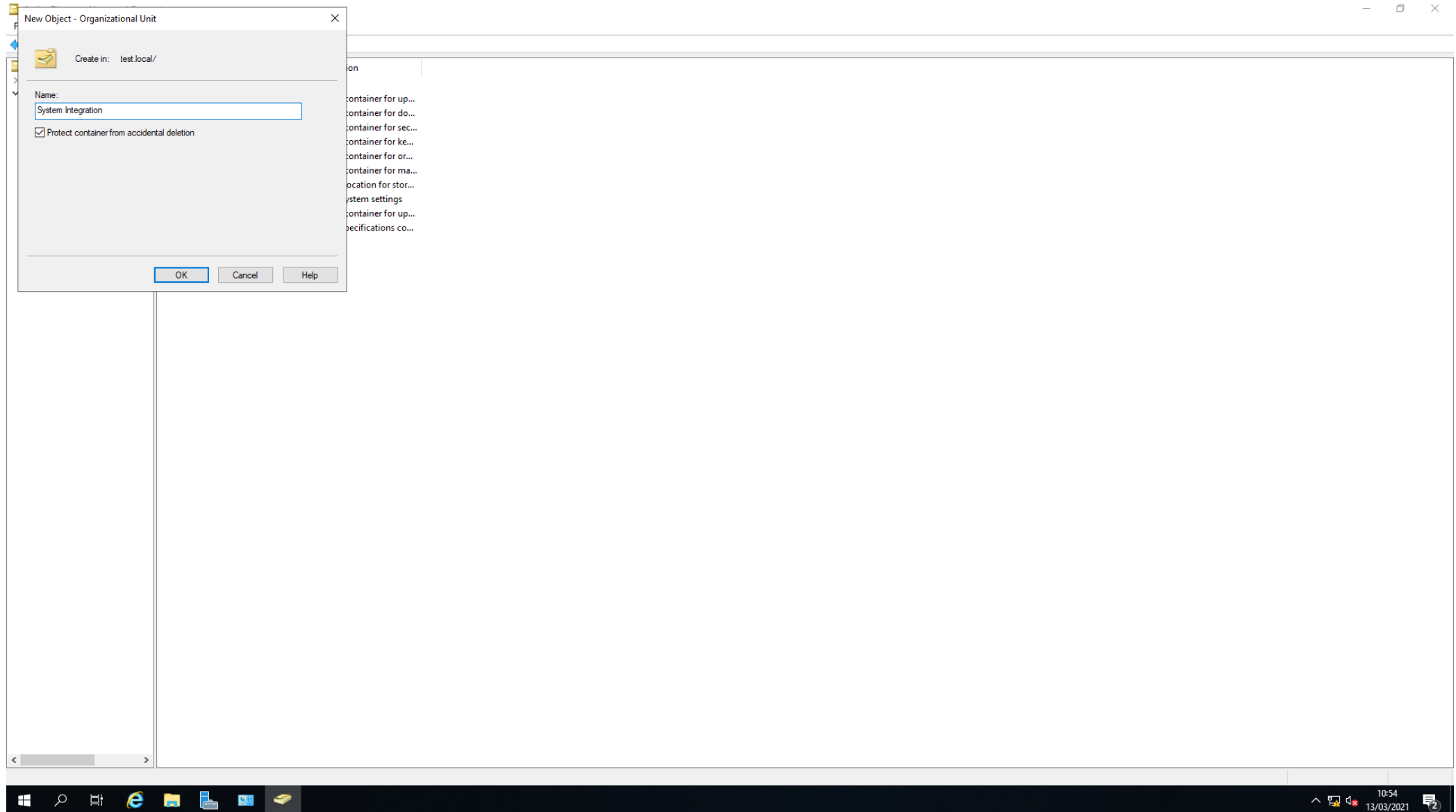
Active Directory Users and Computers



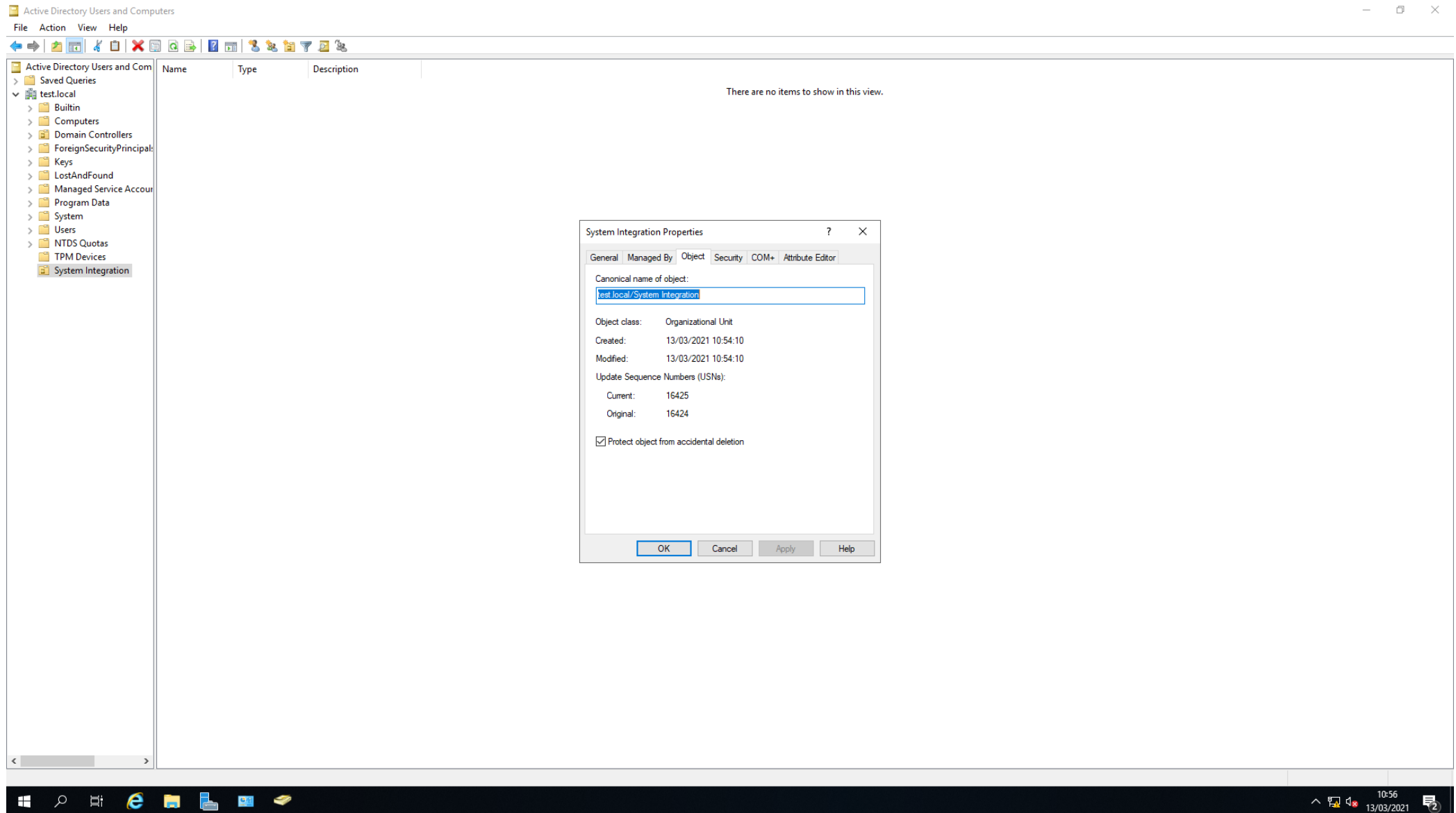
Active Directory Users and Computers



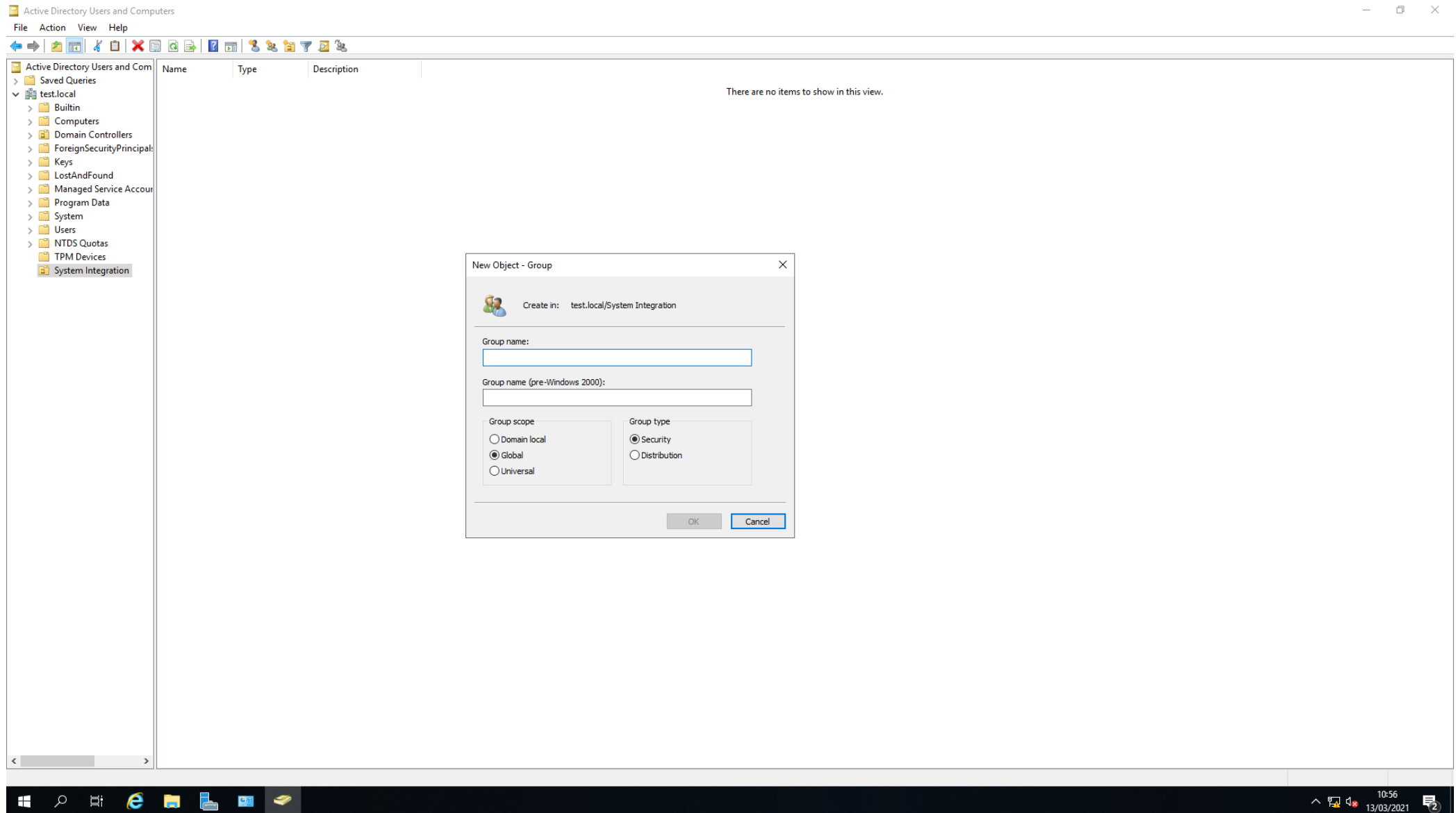
Active Directory Users and Computers



Active Directory Users and Computers



Active Directory Users and Computers

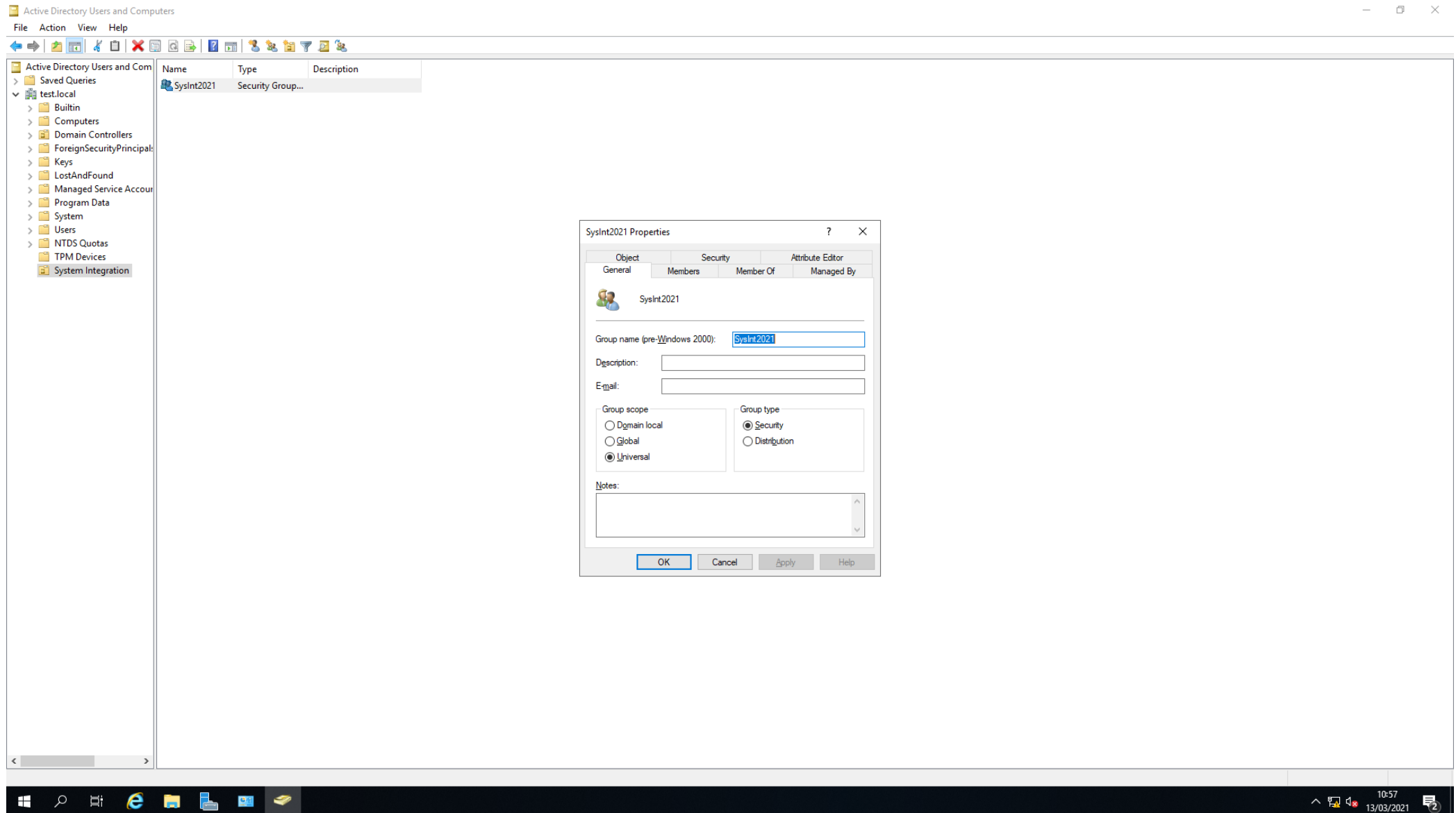


Active Directory Users and Computers

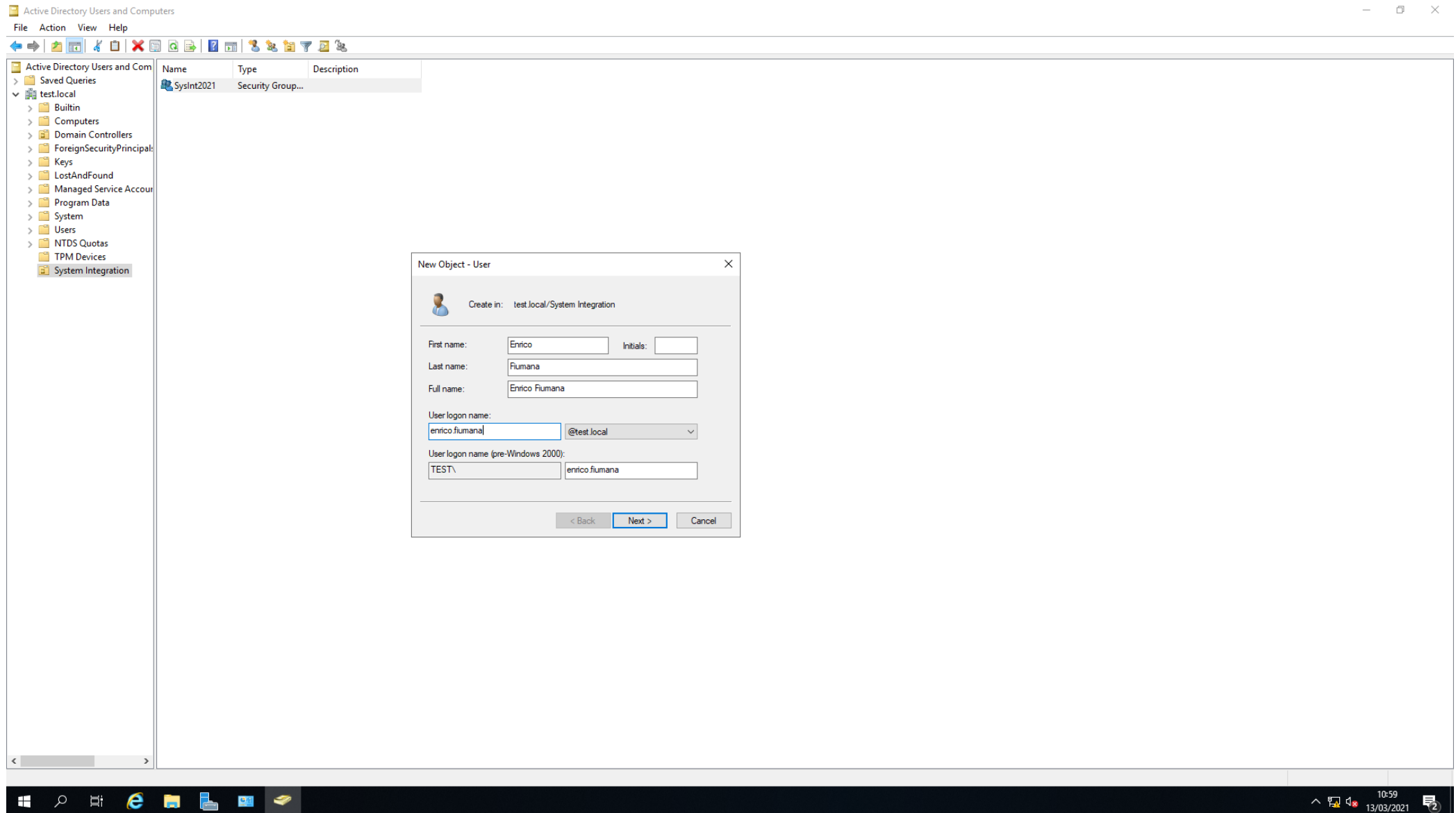
Group scope e Group type

- Scope
 - **Domain local**: visibili solo all'interno del dominio. Possono contenere solo gruppi Domain local ma oggetti sia del dominio che dei domini in trust
 - **Global**: visibili in tutta la foresta, possono contenere solo oggetti del dominio in cui viene creato.
 - **Universal**: visibili in tutta la foresta, possono contenere oggetti di tutta la foresta
- Types
 - **Security**: utilizzati per fornire accessi ad oggetti
 - **Distribution**: utilizzati per creare liste di distribuzione via mail

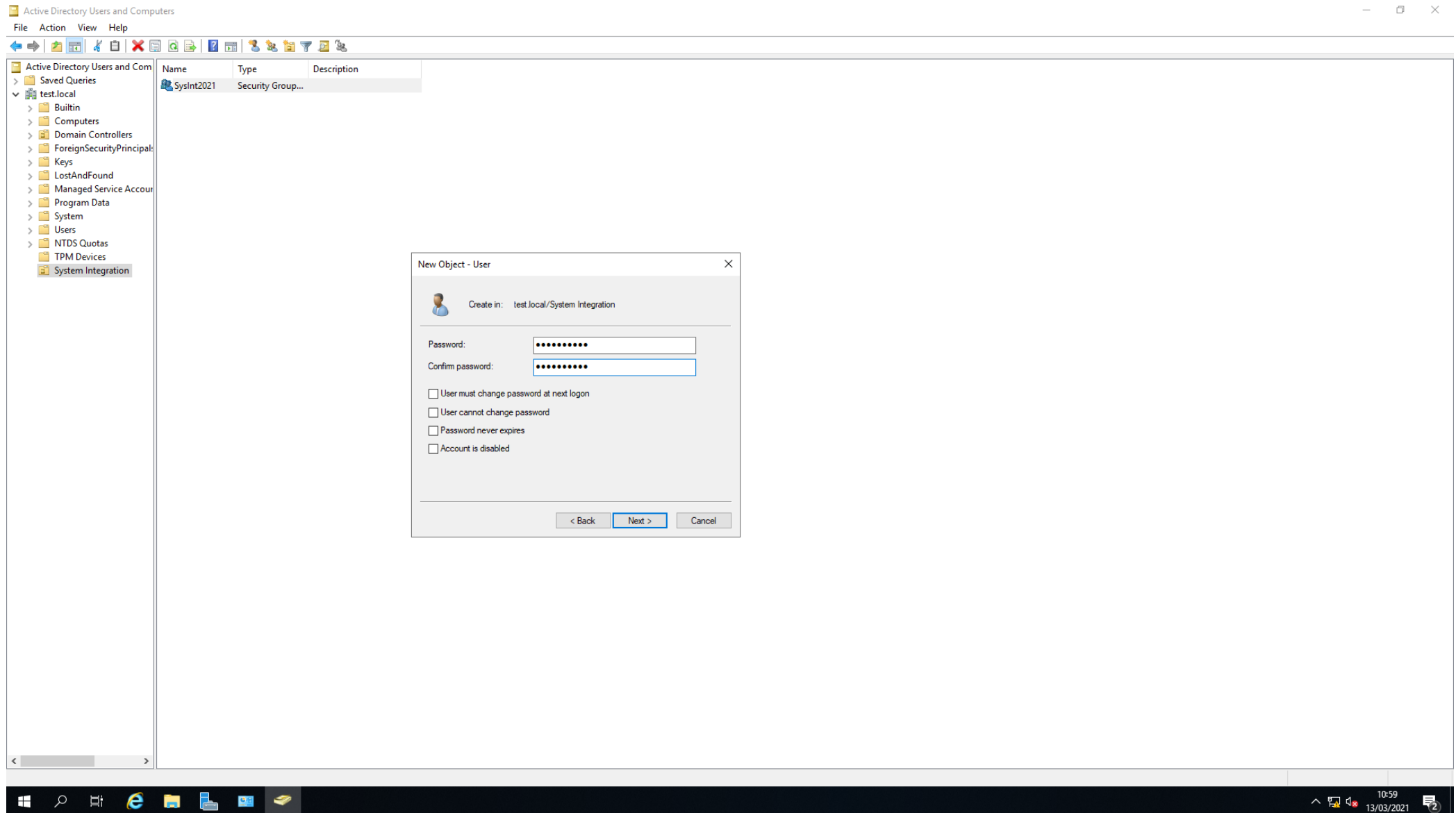
Active Directory Users and Computers



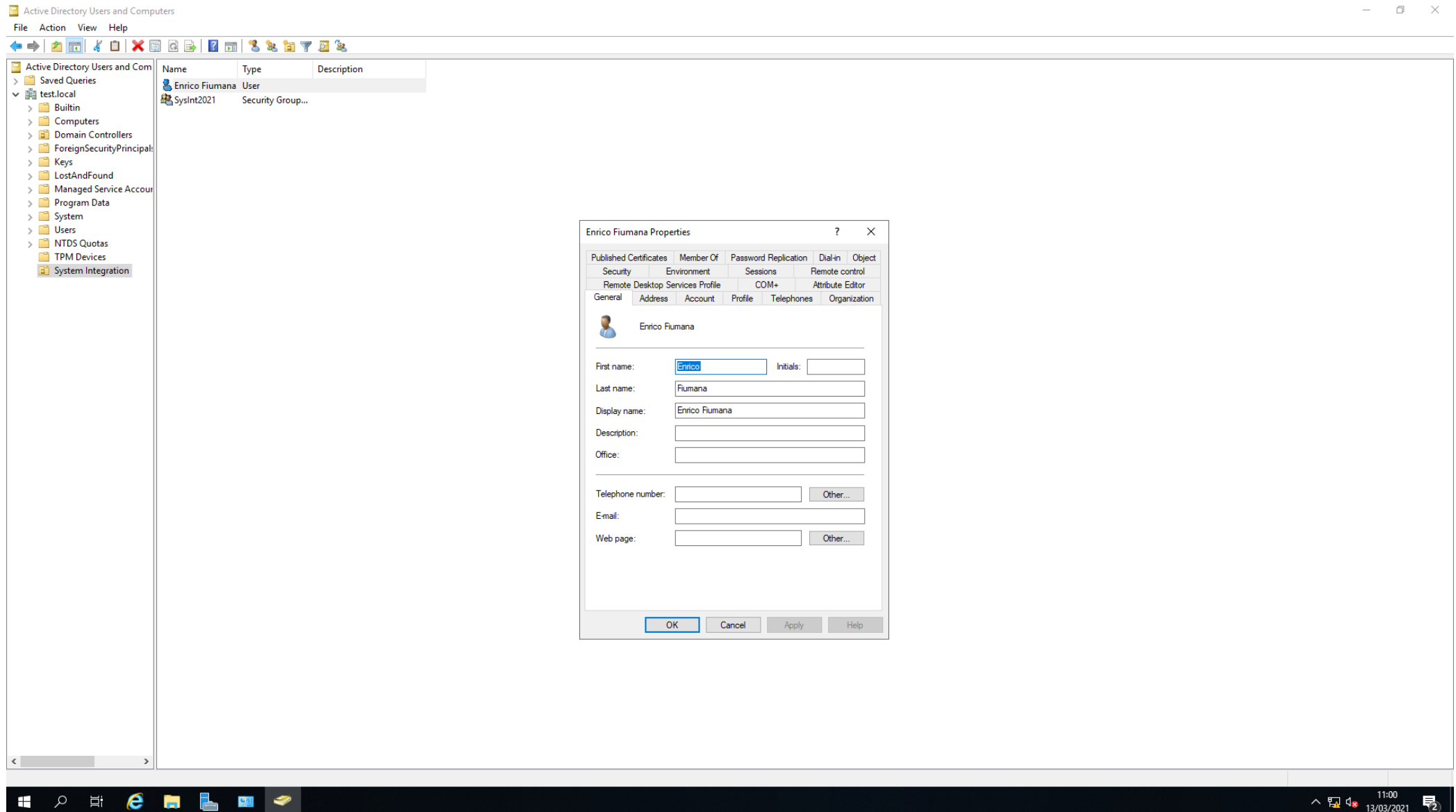
Active Directory Users and Computers



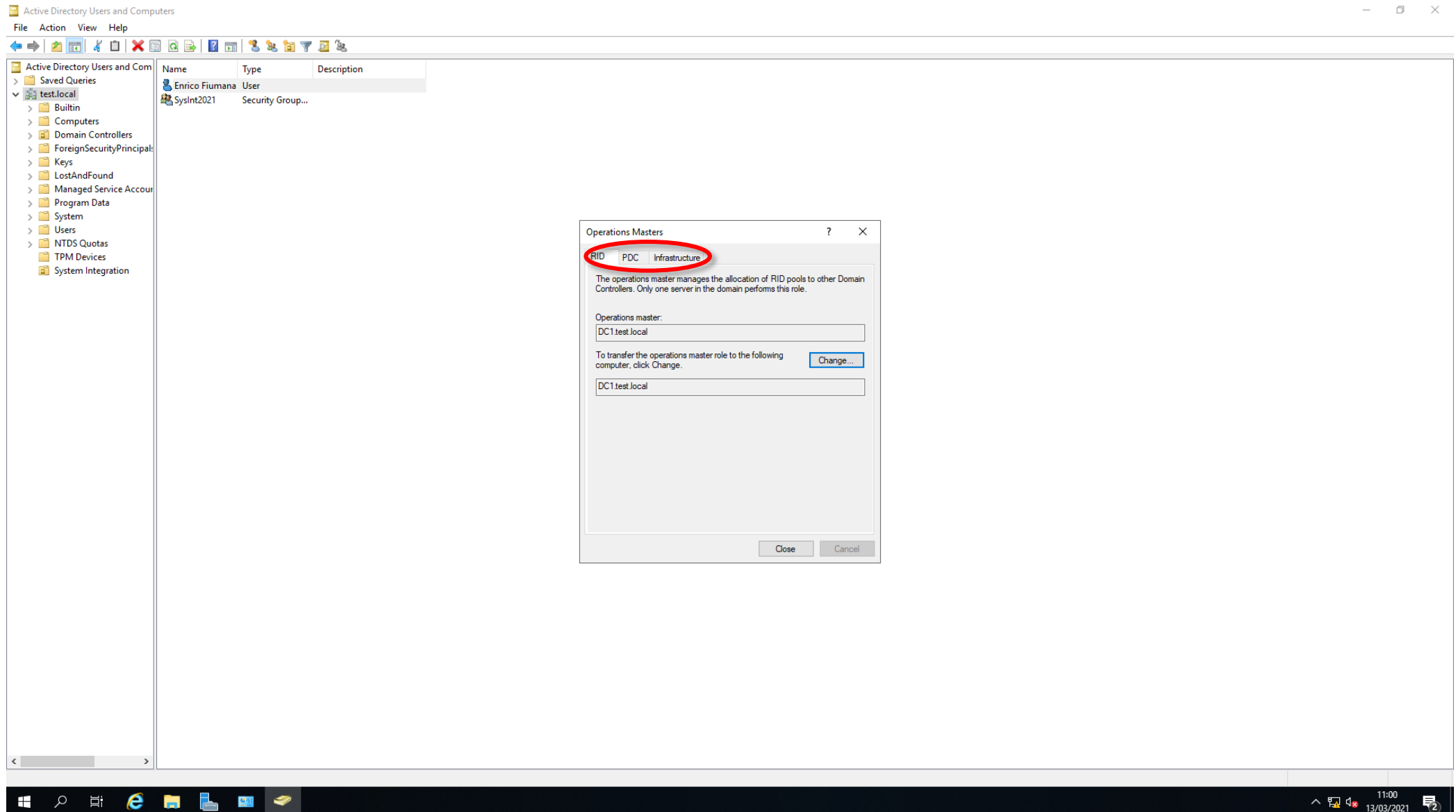
Active Directory Users and Computers



Active Directory Users and Computers



Active Directory Users and Computers



Active Directory Sites and Services

- Definiamo i siti FISICI dei vari domini di una foresta con i costi di collegamento
- Definiamo quali sono i DC con il global catalog

Active Directory Sites and Services

The screenshot shows the Active Directory Sites and Services console. The left pane displays the hierarchy: Active Directory Sites and Services > Sites > Inter-Site Transports > Subnets > Default-First-Site-Name > Servers > DC1 > NTDS Settings. The main pane is empty, displaying the message "There are no items to show in this view." A dialog box titled "NTDS Settings Properties" is open, showing the "General" tab. The "Description" field is empty. The "Query Policy" dropdown is set to "Default". The "DNS Alias" field contains the value "91C42EF2-23C2-49DD-88AE-18CDD935B040_msdcs.t". The "Global Catalog" checkbox is checked and circled in red. Below the checkbox, a note states: "The amount of time it will take to publish the Global Catalog varies depending on your replication topology." The dialog box has "OK", "Cancel", "Apply", and "Help" buttons at the bottom.

Active Directory Sites and Services

File Action View Help

Active Directory Sites and Services

Sites

- Inter-Site Transports
- Subnets
- Default-First-Site-Name
- Servers
- DC1
- NTDS Settings

Name	From Server	From Site	Type	Description
There are no items to show in this view.				

NTDS Settings Properties

General Connections Object Security Attribute Editor

NTDS Settings

Description:

Query Policy:

DNS Alias: 91C42EF2-23C2-49DD-88AE-18CDD935B040_msdcs.t

☒ Global Catalog

The amount of time it will take to publish the Global Catalog varies depending on your replication topology.

OK Cancel Apply Help

11:01 13/03/2021

Active Directory Sites and Services

Active Directory Sites and Services [CSI-DSA1.campusfc.dir.unit]

File Action View Help

Active Directory Sites and Services [CSI-DSA1.campusfc.dir.unit]

Sites

- Inter-Site Transports
 - IP
 - SMTP
- Subnets
 - 10.0.0.0/8
 - 10.190.0.0/24
 - 10.192.0.0/16
 - 10.200.0.0/22
 - 10.21.0.0/16
 - 10.221.0.0/16
 - 10.222.0.0/16
 - 10.254.250.0/24
 - 10.41.0.0/16
 - 130.136.0.0/16
 - 137.204.0.0/16
 - 137.204.107.0/24
 - 137.204.11.0/24
 - 137.204.115.0/24
 - 137.204.116.0/24
 - 137.204.14.0/24
 - 137.204.153.0/24
 - 137.204.182.0/24
 - 137.204.183.0/24
 - 137.204.191.0/24
 - 137.204.202.0/24
 - 137.204.210.0/24
 - 137.204.211.0/24
 - 137.204.43.0/24
 - 137.204.67.0/24
 - 137.204.72.0/24
 - 137.204.74.0/24
 - 137.204.75.0/24
 - 137.204.78.0/24
 - 137.204.79.0/24
 - 137.204.85.0/24
 - 137.204.86.0/24
 - 137.204.87.0/24
 - 137.204.9.0/24
 - 172.16.0.0/12
 - 172.30.200.0/24
 - 192.12.47.0/24
 - 192.167.52.0/22
 - 192.168.0.0/16
 - 193.205.150.0/23
 - 193.43.97.0/24
 - CESIA-AZURE
 - CESIA-BO
 - Isolation
 - POLOCESENA

Name	Type	Description	Cost	Replication Interval
CESIA-BO-CESIA-AZURE	Site Link		150	15
CESIA-BO-Isolation	Site Link		100	15
CESIA-BO-POLOCESENA	Site Link		100	15
CESIA-BO-POLORIMINI	Site Link		100	15

Windows taskbar: 4:00 PM 3/15/2021

Active Directory Sites and Services

Active Directory Sites and Services [CSI-DSA1.campusfc.dir.unibo.it]

File Action View Help

Active Directory Sites and Services [CSI-DSA1.campusfc.dir.unibo.it]

Sites

- Inter-Site Transports
- Subnets
- CESIA-AZURE
 - Servers
 - DCDIR-AZ0
 - DCDIR-AZ1
 - DCPERS-AZ0
 - DCPERS-AZ1
 - DCSTUD-AZ0
 - DCSTUD-AZ1
- CESIA-BO
 - Servers
 - AKI
 - ATU
 - DC03
 - DC04
 - DCRDIR
 - DCRDIR02
 - DCRHQ
 - DCRHQ02
 - DCRPERSONALE01
 - DCRPERSONALE02
 - DCRSTUDENTI1
 - DCRSTUDENTI02
 - EDGE
 - POLE
 - RADI
 - RAZOR
- Isolation
- POLOCESENA
 - Servers
 - CSI-DSA1
 - CSI-DSA2
 - CSI-DSA3
- POLORIMINI
 - Servers
 - DSA1RIMINI
 - DSA3RIMINI
 - RN-DSA1-2019

Name	Type	Description
IP	Inter-Site Trans...	
SMTP	Inter-Site Trans...	

Windows taskbar: 4:00 PM 3/15/2021

DNS

- Gestiamo il DNS
- Il DNS Microsoft può essere configurato come DDNS – Dynamic DNS
- Possiamo creare e modificare diversi server DNS
- Possiamo creare e modificare altre zone
- Possiamo gestire eventuali server secondari
- Da qui inseriamo gli eventuali forwarders
- Inseriamo, modifichiamo e cancelliamo i RR del DNS
- Effettuiamo lo scavenging delle zone

DNS

DNS Manager

File Action View Help

DNS

DC1

Forward Lookup Zones

test.local

Reverse Lookup Zones

Trust Points

Conditional Forwarders

Name

Type

Data

Timestamp

Forward Lookup Zones

test.local

Reverse Lookup Zones

Trust Points

Conditional Forwarders

DC1

Forward Lookup Zones

test.local

Reverse Lookup Zones

Trust Points

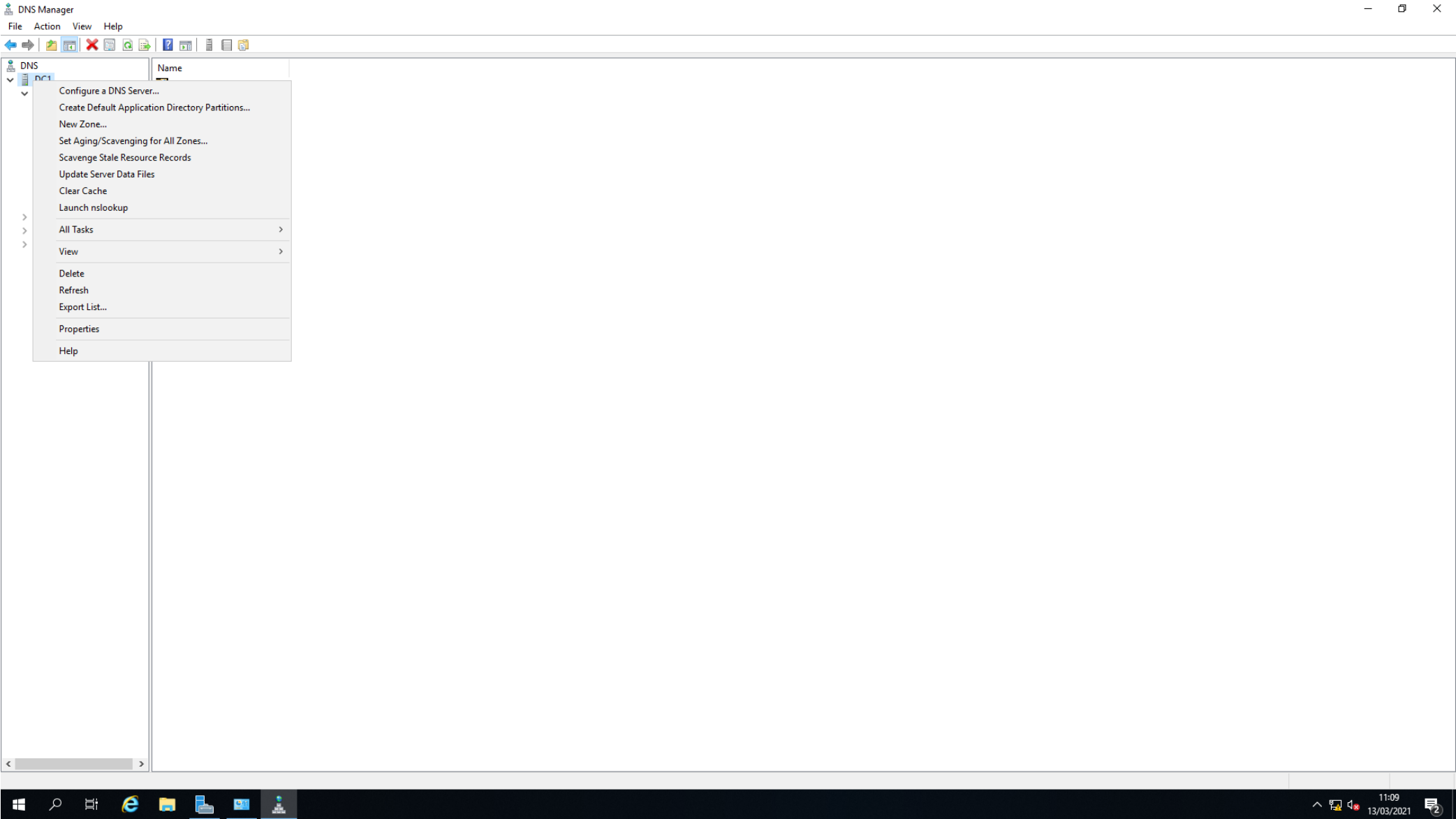
Conditional Forwarders

Name	Type	Data	Timestamp	
Start of Authority (SOA)		[19], dc1.test.local, hostm...	static	
Name Server (NS)		dc1.test.local.	static	
Host (A)		192.168.100.10	06/03/2021 18:00:00	
Host (A)		192.168.100.10	static	

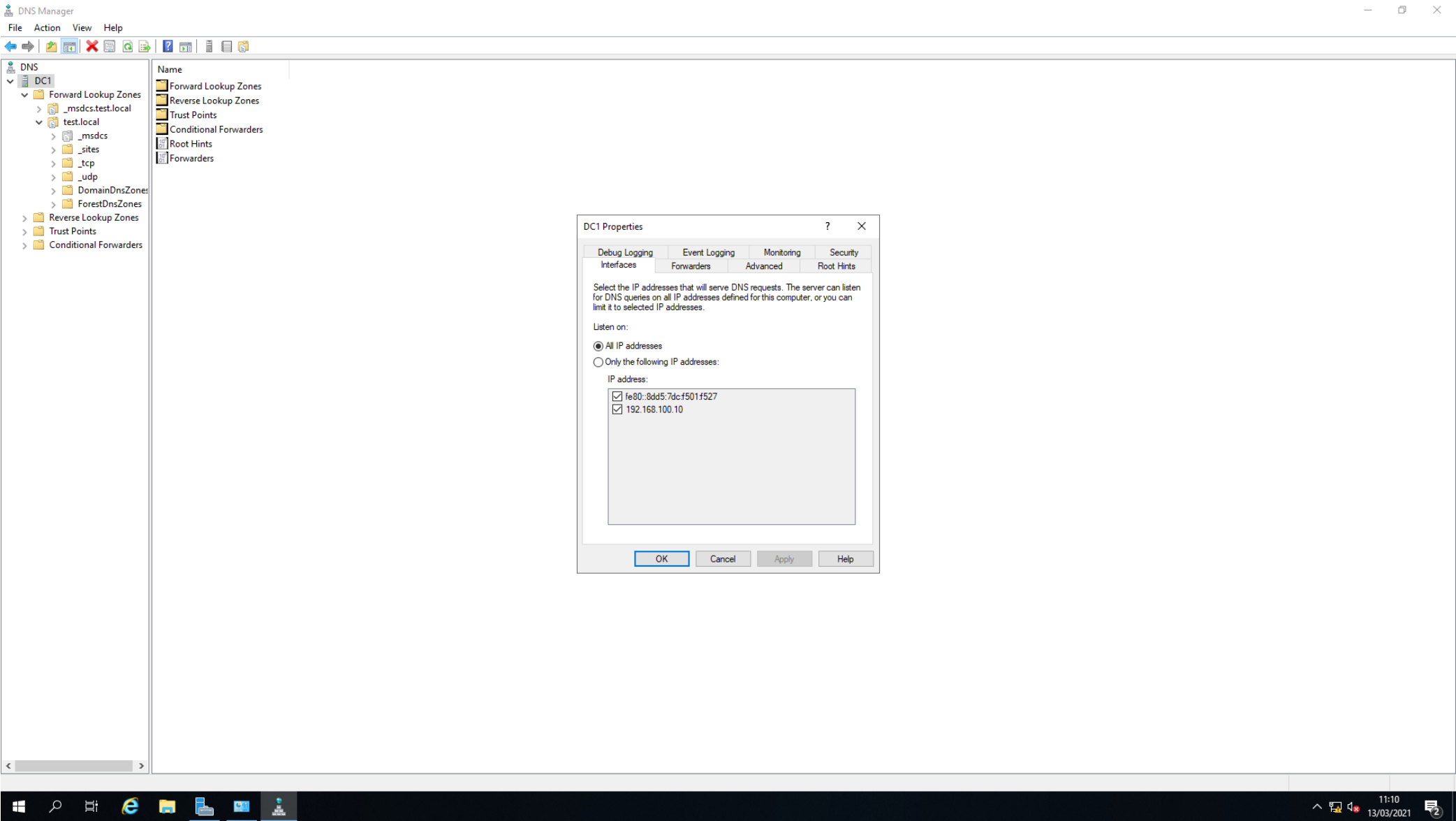
11:09

13/03/2021

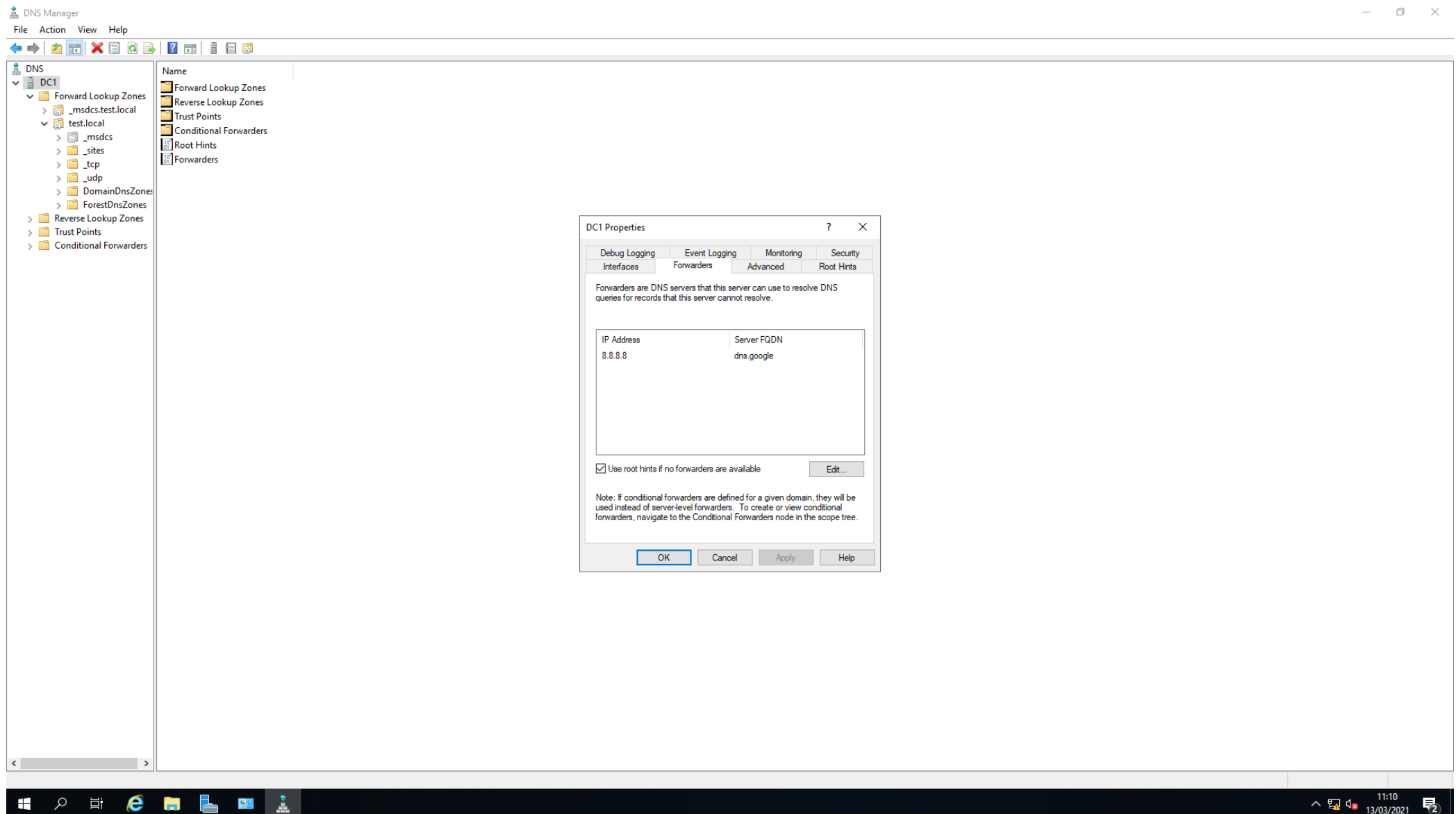
DNS



DNS



DNS



DNS

The screenshot shows the Windows DNS Manager console on the left and the 'DC1 Properties' dialog box on the right. The console displays a tree view of DNS zones under 'DC1', including Forward Lookup Zones, Reverse Lookup Zones, Trust Points, Conditional Forwarders, Root Hints, and Forwarders. The 'DC1 Properties' dialog box is open to the 'Root Hints' tab, showing a list of root servers and their IP addresses. The 'a.root-servers.net' entry is selected.

DNS Manager Console:

- DNS
 - DC1
 - Forward Lookup Zones
 - _msdcs.test.local
 - test.local
 - _msdcs
 - _sites
 - _tcp
 - _udp
 - DomainDnsZones
 - ForestDnsZones
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

DC1 Properties - Root Hints Tab:

Root hints resolve queries for zones that do not exist on the local DNS server. They are only used if forwarders are not configured or fail to respond.

Name servers:

Server Fully Qualified Domain Name (FQDN)	IP Address
a.root-servers.net.	[2001:503:ba3e::2]
b.root-servers.net.	[2001:500:200::b]
c.root-servers.net.	[2001:500:2::c]
d.root-servers.net.	[2001:500:2d::d]
e.root-servers.net.	[2001:500:a8::e]
f.root-servers.net.	[2001:500:2f::f]
g.root-servers.net.	[2001:500:12::d0d]
h.root-servers.net.	[2001:500:1::53]

Buttons: Add..., Edit..., Remove, Copy from Server, OK, Cancel, Apply, Help.

DNS

DNS Manager

File Action View Help

DNS

- DC1
 - Forward Lookup Zones
 - _msdcs.test.local
 - test.local
 - _msdcs
 - _sites
 - _tcp
 - _udp
 - DomainDnsZones
 - ForestDnsZones
 - (same as parent folder)
 - (same as parent folder)
 - (same as parent folder)
 - dc1
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[19], dc1.test.local, hostm...	static
(same as parent folder)	Name Server (NS)	dc1.test.local.	static
(same as parent folder)	Host (A)	192.168.100.10	06/03/2021 18:00:00
(same as parent folder)	Host (A)	192.168.100.10	static

test.local Properties

WINS Zone Transfers Security

General Start of Authority (SOA) Name Servers

Status: Running

Type: Active Directory-Integrated

Replication: All DNS servers in this domain

Data is stored in Active Directory.

Dynamic updates: Secure only

 Allowing nonsecure dynamic updates is a significant security vulnerability because updates can be accepted from untrusted sources.

To set aging/scavenging properties, click Aging...

DNS

DNS Manager

File Action View Help

DNS

- DC1
 - Forward Lookup Zones
 - _msdcs.test.local
 - test.local
 - _msdcs
 - _sites
 - _tcp
 - _udp
 - DomainDnsZones
 - ForestDnsZones
 - (same as parent folder)
 - (same as parent folder)
 - (same as parent folder)
 - (same as parent folder)
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[19], dc1.test.local, hostm...	static
(same as parent folder)	Name Server (NS)	dc1.test.local.	static
(same as parent folder)	Host (A)	192.168.100.10	06/03/2021 18:00:00
(same as parent folder)	Host (A)	192.168.100.10	static
dc1			

test.local Properties

WINS Zone Transfers Security

General Start of Authority (SOA) Name Servers

Serial number: 10 Increment

Primary server: dc1.test.local. Browse...

Responsible person: hostmaster.test.local. Browse...

Refresh interval: 15 minutes

Retry interval: 10 minutes

Expires after: 1 days

Minimum (default) TTL: 1 hours

TTL for this record: 0 :1 :0 :0 (DDDD:HH:MM:SS)

OK Cancel Apply Help

11:11 13/03/2021

DNS

DNS Manager

File Action View Help

DNS

DC1

Forward Lookup Zones

Reverse Lookup Zones

Trust Points

Conditional Forwarders

_msdcs

_sites

_tcp

_udp

DomainDnsZones

ForestDnsZones

(same as parent folder)

(same as parent folder)

(same as parent folder)

dc1

Name

Type

Data

Timestamp

Start of Authority (SOA)

Name Server (NS)

Host (A)

Host (A)

[21], dc1.test.local

dc1.test.local

192.168.100.10

192.168.100.10

test.local Properties

WINS

Zone Transfers

Security

General

Start of Authority (SOA)

Name Servers

To add name servers to the list, click Add.

Name servers:

Server Fully Qualified Domain Name (FQDN)

IP Address

dc1.test.local

[192.168.100.10]

Add...

Edit...

Remove

* represents an IP address retrieved as the result of a DNS query and may not represent actual records stored on this server.

OK

Cancel

Apply

Help

16:19

15/03/2021

DNS

DNS Manager

File Action View Help

DNS

- DC1
 - Forward Lookup Zones
 - _msdcs.test.local
 - test.local
 - _msdcs
 - _sites
 - _tcp
 - _udp
 - DomainDnsZones
 - ForestDnsZones
 - (same as parent folder)
 - (same as parent folder)
 - (same as parent folder)
 - dc1
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

Name	Type	Data	Timestamp
_msdcs			
_sites			
_tcp			
_udp			
DomainDnsZones			
ForestDnsZones			
(same as parent folder)	Start of Authority (SOA)	[19], dc1.test.local, hostm...	static
(same as parent folder)	Name Server (NS)	dc1.test.local.	static
(same as parent folder)	Host (A)	192.168.100.10	06/03/2021 18:00:00
(same as parent folder)	Host (A)	192.168.100.10	static

test.local Properties

General Start of Authority (SOA) Name Servers

WINS Zone Transfers Security

A zone transfer sends a copy of the zone to the servers that request a copy.

☐ Allow zone transfers:

- ☒ To any server
- ☐ Only to servers listed on the Name Servers tab
- ☐ Only to the following servers

IP Address	Server FQDN
------------	-------------

Edit

To specify secondary servers to be notified of zone updates, click Notify.

Notify...

OK Cancel Apply Help

11:12 13/03/2021

Active Directory Schema

Fino ad ora abbiamo visto come trasferire tutti i ruoli FSMO a parte quello dello schema master.

Per gestire questo ruolo è necessario attivare lo strumento «Active Directory Schema» che di default non viene visualizzato fra gli strumenti disponibili.

Per attivarlo bisogna, da un prompt dei comandi aperto con diritti amministrativi, dare il seguente comando:

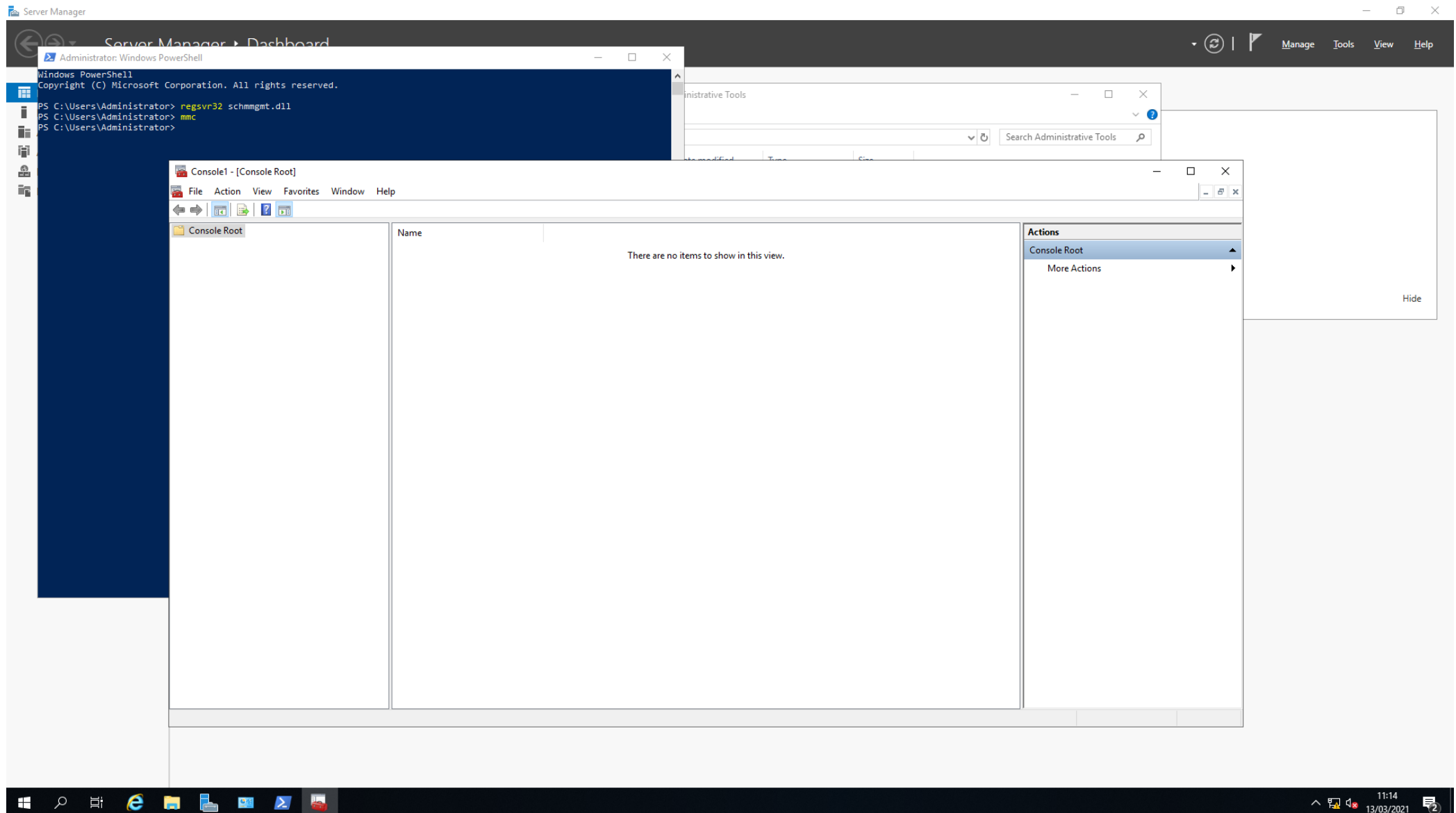
Regsvr32 schmmgmt.dll

A questo punto sarà possibile aggiungere il modulo (snap-in)

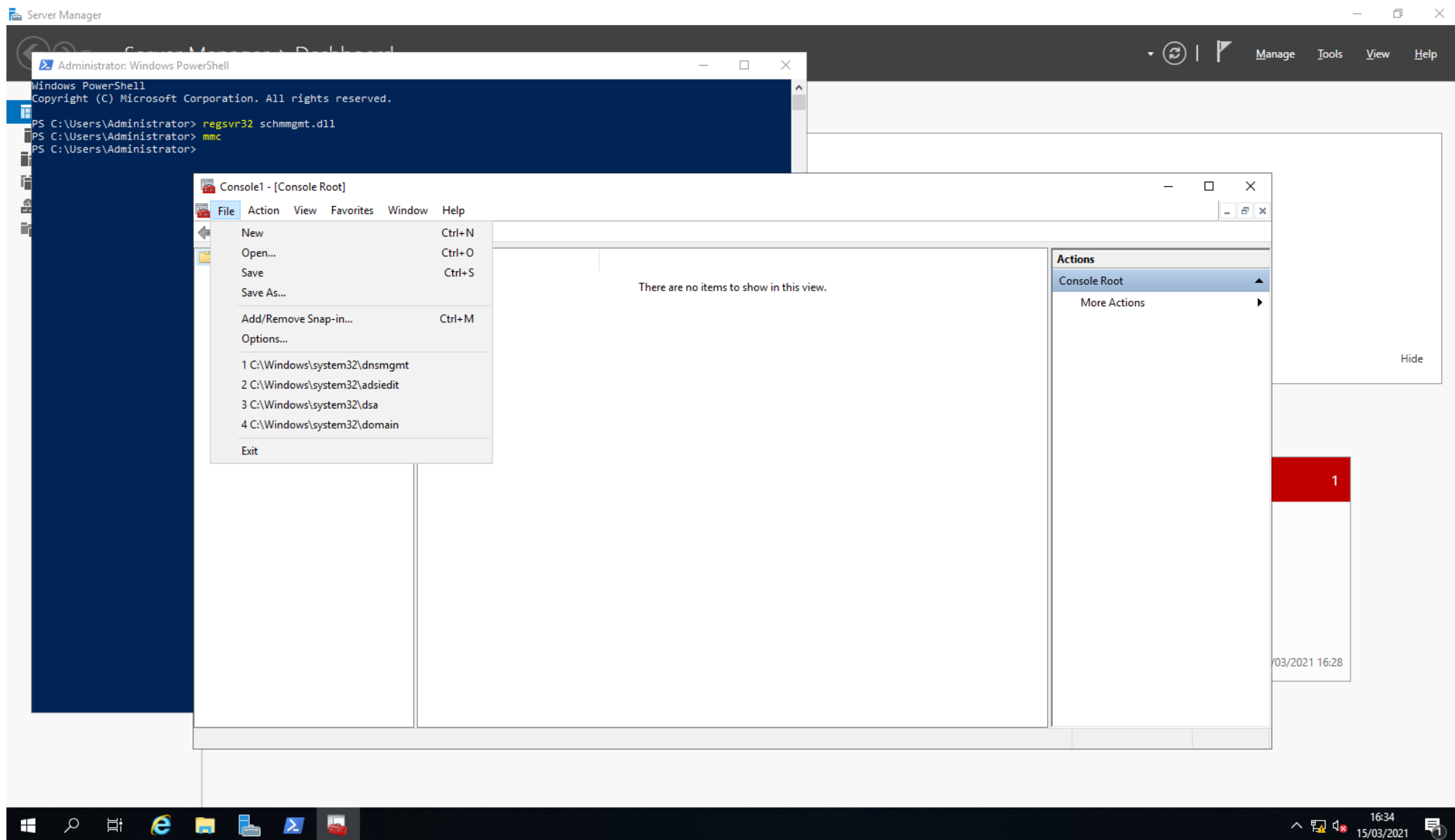
Active Directory Schema

alla console di gestione

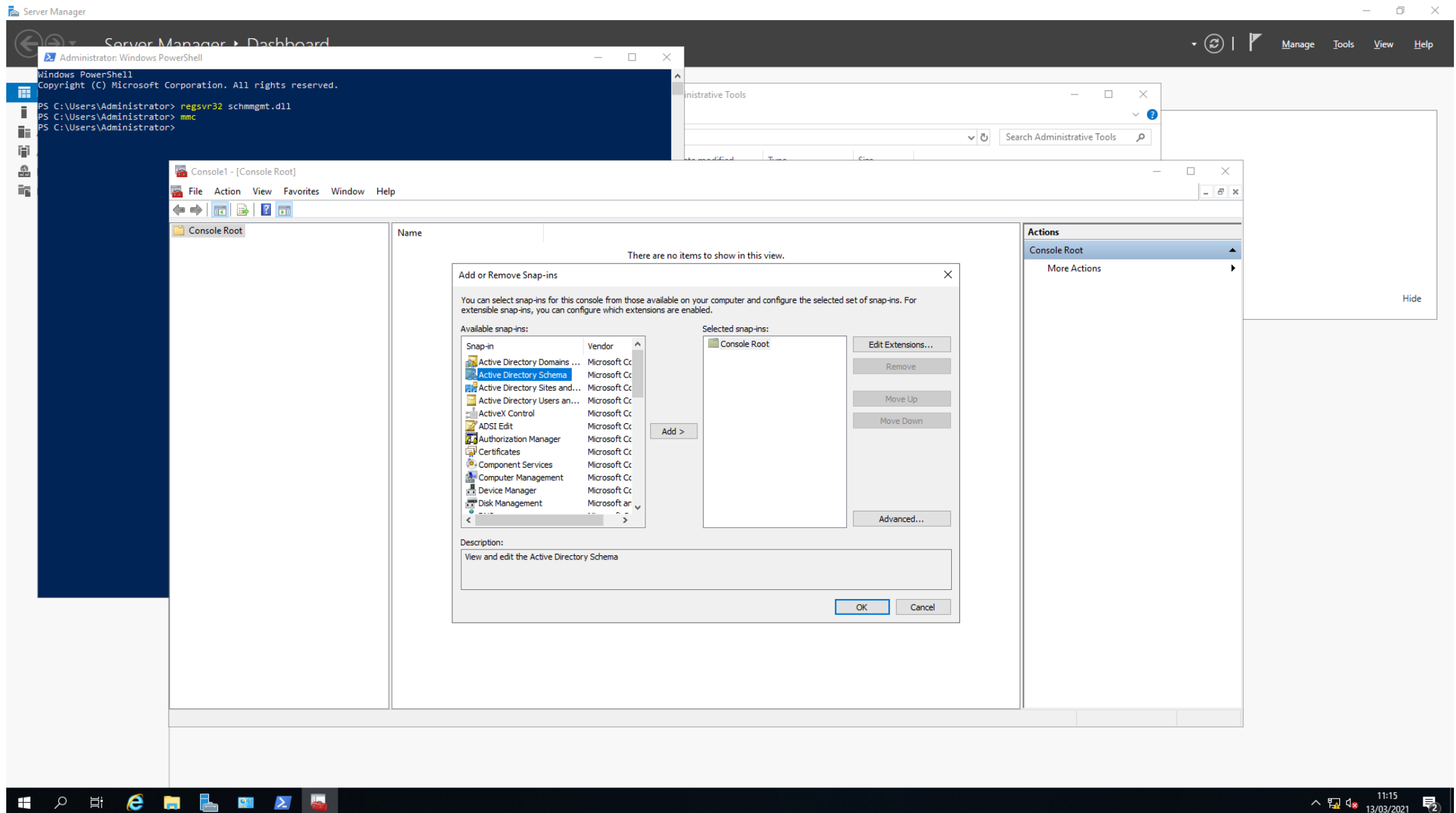
Active Directory Schema



Active Directory Schema



Active Directory Schema



Active Directory Schema

The screenshot displays the Windows Server Manager interface. In the foreground, a Windows PowerShell window is open, showing the following commands and output:

```
PS C:\Users\Administrator> regsvr32 schmmgmt.dll
PS C:\Users\Administrator> mmc
PS C:\Users\Administrator>
```

Below the PowerShell window, the 'Console1 - [Console Root\Active Directory Schema [DC1.test.local]\Classes]' console is open. It shows a tree view of the Active Directory Schema with the 'Classes' folder expanded. The main pane displays a table of classes:

Name	Type	Status	Description
account	Structural	Active	The account object clas...
aCSPolicy	Structural	Active	ACS-Policy
aCSResourceLimits	Structural	Active	ACS-Resource-Limits
aCSSubnet	Structural	Active	ACS-Subnet
addressBookContainer	Structural	Active	Address-Book-Container
addressTemplate	Structural	Active	Address-Template
applicationEntity	Structural	Active	Application-Entity
applicationProcess	Structural	Active	Application-Process
applicationSettings	Abstract	Active	Application-Settings
applicationSiteSettings	Abstract	Active	Application-Site-Settings
applicationVersion	Structural	Active	Stores versioning inform...
attributeSchema	Structural	Active	Attribute-Schema
bootableDevice	Auxiliary	Active	A device with boot para...
builtinDomain	Structural	Active	Builtin-Domain
categoryRegistration	Structural	Active	Category-Registration
certificationAuthority	Type 88	Active	Certification-Authority
classRegistration	Structural	Active	Class-Registration
classSchema	Structural	Active	Class-Schema
classStore	Structural	Active	Class-Store
comConnectionPoint	Structural	Active	Com-Connection-Point
computer	Structural	Active	Computer
configuration	Structural	Active	Configuration
connectionPoint	Abstract	Active	Connection-Point
contact	Structural	Active	Contact
container	Structural	Active	Container
controlAccessRight	Structural	Active	Control-Access-Right
country	Type 88	Active	Country
cRLDistributionPoint	Structural	Active	CRL-Distribution-Point
crossRef	Structural	Active	Cross-Ref
crossRefContainer	Structural	Active	Cross-Ref-Container
device	Type 88	Active	Device
dfsConfiguration	Structural	Active	Dfs-Configuration

The right-hand pane of the console shows the 'Actions' menu with 'Classes' selected and 'More Actions' available. The bottom of the screen shows the Windows taskbar with the time 11:17 and date 13/03/2021.

Active Directory Schema

The screenshot displays the Windows Server Manager interface. In the background, a Windows PowerShell window shows the following commands and output:

```
PS C:\Users\Administrator> regsvr32 schmmgmt.dll
PS C:\Users\Administrator> mmc
PS C:\Users\Administrator>
```

The foreground window is the 'Console1 - [Console Root\Active Directory Schema [DC1.test.local]\Classes]' console. A context menu is open over the 'Classes' folder in the left-hand tree view. The menu options are:

- Change Active Directory Domain Controller...
- Connect to Schema Operations Master
- Operations Master... (highlighted)
- Permissions...
- Reload the Schema
- New Window from Here
- Refresh
- Help

The main pane of the console displays a table of Active Directory Schema classes:

Name	Type	Status	Description
Account-Object-Class	Structural	Active	The account object class
ACS-Policy	Structural	Active	ACS-Policy
ACS-Resource-Limits	Structural	Active	ACS-Resource-Limits
ACS-Subnet	Structural	Active	ACS-Subnet
Address-Book-Container	Structural	Active	Address-Book-Container
Address-Template	Structural	Active	Address-Template
Application-Entity	Structural	Active	Application-Entity
Application-Process	Structural	Active	Application-Process
Application-Settings	Structural	Active	Application-Settings
Application-Site-Settings	Structural	Active	Application-Site-Settings
Stores versioning information	Structural	Active	Stores versioning information
Attribute-Schema	Structural	Active	Attribute-Schema
A device with boot parameters	Auxiliary	Active	A device with boot parameters
Builtin-Domain	Structural	Active	Builtin-Domain
Category-Registration	Structural	Active	Category-Registration
Certification-Authority	Type 88	Active	Certification-Authority
Class-Registration	Structural	Active	Class-Registration
Class-Schema	Structural	Active	Class-Schema
Class-Store	Structural	Active	Class-Store
Com-Connection-Point	Structural	Active	Com-Connection-Point
Computer	Structural	Active	Computer
Configuration	Structural	Active	Configuration
Connection-Point	Abstract	Active	Connection-Point
Contact	Structural	Active	Contact
Container	Structural	Active	Container
Control-Access-Right	Structural	Active	Control-Access-Right
Country	Type 88	Active	Country
CRL-Distribution-Point	Structural	Active	CRL-Distribution-Point
Cross-Ref	Structural	Active	Cross-Ref
Cross-Ref-Container	Structural	Active	Cross-Ref-Container
Device	Type 88	Active	Device
Dfs-Configuration	Structural	Active	Dfs-Configuration
DNIS-Class	Structural	Active	DNIS-Class

The bottom of the console shows the text 'Change the Operations Master'.